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1) $A(3,0)$ e $B(0,-4)$

Eq $\begin{cases} x = x_1 + at \\ y = y_1 + bt \\ z = z_1 + ct \end{cases}$

$\downarrow$

como é $\mathbb{R}^2$

$\begin{cases} x = x_1 + at \\ y = y_1 + bt \end{cases}$

$ax + by + c = 0$

$\downarrow$

$\vec{AB}$ calcular o Vetor $\vec{AB}$

$\vec{AB} = B - A = (0,4) - (3,0)$

$\vec{AB} = (-3,4)$

Eq. Vetorial

$(x, y, z) = (x_1, y_1, z_1) + t(a,b,c)$

$\downarrow$

$\mathbb{R}^2$

$\downarrow$

$(x, y) = (3,0) + t(-3,4)$

Paramétrica:

$x = 2 - 3t$

$y = 0 + 4t$

$y = 4t$

Assim:

Eq Paramétrica pertencente a A $\begin{cases} x = 2 - 3t \\ y = 4t \end{cases}$

como A e B são paramétricas e cada uma possui duas paramétrica e duas vetorial e duas cartesiana

Agora B

$(x, y) = (0, -4) + t(-3, 4)$

$x = 0 + (-3)t$

$x = -3t$

$y = -4 + 4$

$\downarrow$

Eq Paramétrica B $\begin{cases} x = -3t \\ y = 0 \end{cases}$

Eq. Vetorial A:

$(x, y) = (3, 0) + t(-3, 4)$

Eq paramétrica A:

$\begin{cases} x = 2 - 3t \\ y = 4t \end{cases}$

Eq. Cartesiana A:

$ax + by + c = 0$

$\vec{AB} = (-3, 4)$

$\vec{m} = (4, 3)$

$A(3, 0)$

$4 \cdot x + 3y + c = 0$

$4 \cdot 3 + \cancel{3}^{0} \cdot 0 + c = 0$

$12 + 0 + c = 0$

$12 + c = 0$

$c = -12$

Eq. Vetorial B:

$(x, y) = (0, -4) + t(-3, 4)$

Eq. paramétrica B:

$\begin{cases} x = -3t \\ y = 0 \end{cases}$

Assim:

$4x + 3y - 12 = 0$ A

Eq. Cartesiana B

$\overrightarrow{AB} = (-3, 4)$ $\qquad$ $B = (0, -4)$

$\vec{m} = (4, 3)$

$ax + by + c = 0$

$4 \cdot \overset{0}{0} + 3 \cdot (-4) + c = 0$

$-12 + c = 0$

$c = 12$

Assim:

$\underline{4x + 3\overset{B}{y} + 12 = 0}$

2) $\begin{cases} x = 3 - 2t \\ y = 4 + t \; ; \; t \in \mathbb{R} \end{cases}$

usolar t

$x - 3 = -2t$

$-t = \frac{x-3}{2}$

$t = -\left(\frac{x-3}{2}\right)$

Assim: achar x

$x = 3 - \cancel{2} \cdot \left(-\frac{x-3}{\cancel{2}}\right)$

$x = 3 - x - 3$

$x + x = 3 - 3$

$2x = 0$

$x = \frac{0}{2} = 0$

$t = -\left(\frac{0-3}{2}\right)$

$t = -\left(\frac{-3}{2}\right)$

achar y

$y = 4 + \left(-\left(\frac{-3}{2}\right)\right)$

$y = 4 - \left(\frac{-3}{2}\right)$

$y = 4 + \frac{3}{2}$

$y = \frac{8+3}{2}$ $\quad y = \frac{11}{2}$

Pontos de intersecção:

$\left(0 \text{ e } \frac{11}{2}\right)$

$P = ?$, abscissa é $\frac{1}{2}$

Substituindo na Paramétrica de r

$y = ?$

$$\begin{cases} \frac{1}{2} = 3 - 2t \\ y = 4 + t \end{cases}$$

$1 = 2(3 - 2t)$

$1 = 6 - 4t$

$1 - 6 = -4t$

$-5 = -4t \cdot (-1)$

$4t = 5$

$t = \frac{5}{4}$

achar y

$y = 4 + \frac{5}{4}$

$y = \frac{16+5}{4}$

$y = \frac{21}{4}$

aci

$x = 3 - 2\left(\frac{5}{4}\right)$

$x = -\frac{25}{4}$

$P\left(-\frac{25}{4}, \frac{21}{4}\right)$

3. $r: x + 2y - a = 0$ $\qquad$ $s: 4x + ay - 7 = 0$

$2y = -x + a$ $\qquad$ $ay = -4x + 7$

$y = \frac{-x + a}{2}$ $\qquad$ $y = \frac{-4x + 7}{a}$

$y = \frac{-x}{2} + \frac{a}{2}$ $\qquad$ $y = \frac{-4x}{a} + \frac{7}{a}$

como a reta é perpendicular

$\frac{-1}{2} \cdot \frac{-4}{a} = -1$

$\frac{4}{2a} = -1 \Rightarrow 4 = -1 \cdot 2a$

$\frac{4}{2} = -a$

$\boxed{a = 2}$

Scanned with CamScanner

$\boxed{a = 2}$

4. $r: \sqrt{3}x - 3y - 1 = 0$

$-3y = -\sqrt{3}x + 1$

$-y = \frac{-\sqrt{3}x + 1}{3}$ (-1)

$y = \frac{\sqrt{3}x}{3} - \frac{1}{3}$

$tg\,\theta = \frac{\frac{\sqrt{3}}{3} - 1}{1 + \frac{\sqrt{3}}{3} \cdot 1}$

$tg\,\theta = \frac{\frac{\sqrt{3}}{3} - 1}{1 + \frac{\sqrt{3}}{3}}$

$tg\,\theta = \frac{1 - 1}{1}$

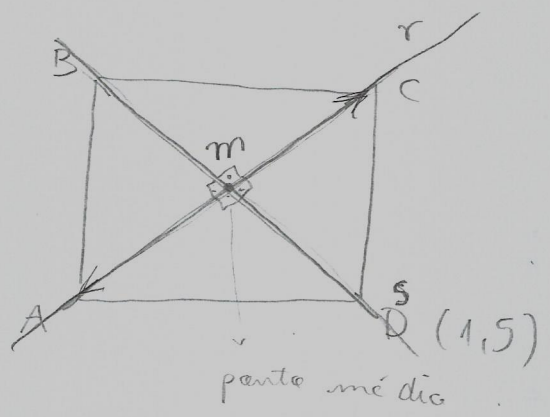
$\boxed{tg\,\theta = 0 = 360°}$

$s: x - 2 = 0$

$y = x + 2$

Scanned with CamScanner

5) $r: 6x - 13y + 2 = 0$ → Reta suporte Diagonal AC



$r \perp s$

$m_r \cdot m_t = -1$

$\overset{a}{6}.x - \overset{b}{13}y + \overset{c}{2} = 0$

$\vec{AC} = (13, 6)$

$\vec{m} = (6, -13)$

$ax + bx + c = 0$

↓

Equação cartesiana

$\vec{AC} = (13, 6)$ $D(1,5)$

$\vec{BD} = ?$

achar coeficiente angular de t

$x_m = \frac{x_A + x_B}{2}$,

$y_m = \frac{y_A + y_B}{2}$

$m_r \cdot m_t = -1$

$m_r = \frac{y_1 - y_2}{x_1 - x_2}$

$m_r = \frac{x_1 - 1}{y_1 - 5}$

$\frac{x_1 - 1}{y_1 - 5} \cdot m_t = -1$

$m_t = \frac{-1}{x_1 - 1}$

$$mt = \frac{-1}{\frac{x_1 - 1}{y_1 - 5}} \qquad mt = -\frac{1}{1} : \frac{y_1 - 5}{x_1 - 1}$$

$$mt = -\frac{(y_1 - 5)}{x_1 - 1}$$

$$M = \left(\frac{x_A + x_B}{2} , \frac{y_A + y_B}{2} \right) \qquad mt = \frac{-y_1 + 5}{x_1 - 1}$$

$$m = \left(\frac{x_A + 1}{2} , \frac{y_A + 5}{2} \right)$$

$$m = \left(\frac{x_A}{2} + \frac{1}{2} , \frac{y_A}{2} + \frac{5}{2} \right)$$

$$\frac{-y_1 + 5}{x_1 - 1} = \frac{y - \frac{y_A + 5}{2}}{x - \frac{x_A + 1}{2}}$$

$$mt = \frac{-1}{\frac{x_1 - 1}{y_1 - 5}} \qquad mt = \frac{-1}{1} : \frac{y_1 - 5}{x_1 - 1}$$

$$mt = \frac{-(y_1 - 5)}{x_1 - 1}$$

$$M = \left(\frac{x_A + x_B}{2} , \frac{y_A + y_B}{2} \right) \qquad mt = \frac{-y_1 + 5}{x_1 - 1}$$

$$m = \left(\frac{x_A + 1}{2} , \frac{y_A + 5}{2} \right)$$

$$m = \left(\frac{x_A}{2} + \frac{1}{2} , \frac{y_A}{2} + \frac{5}{2} \right)$$

$$\frac{-y_1 + 5}{x_1 - 1} = \frac{y - \frac{y_A + 5}{2}}{x - \frac{x_A + 1}{2}}$$